

Subjectivity-Intersection Braid Quantum Gravity

A Constructive Refutation of Objectivist Primacy from Source-Derived Quantum Dynamics to the Einstein Limit

Satoru Watanabe

Subjectivity Intersection Emergence Lab (SIEL)

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Claim status. This is a mathematically explicit *quantum-gravity candidate*, not a completed or empirically confirmed theory. In one fixed pointed-Braid model, the finite source supplies an operational four-direction event cylinder, fixed-model matter routing, the relative factor 3/5, a Lorentz carrier, a source-selected long-wavelength matter action, a unique typed metric solder, Hilbert stress, and an on-shell Ward identity within a declared local, second-order, formally self-adjoint, conservative class. In that same class the common metric variation gives $G_{\mu\nu} = (3/5)T_{\mu\nu}$ and a massless spin-2 low-energy sector with two physical helicities. The unresolved boundary is now the *full finite* Lorentzian quantum parent and its four-component discrete total Ward balance: a temporal total charge is exact, while the three tested spatial environment-only charges obey a range NO-GO. No empirical gravity, ultraviolet completion, calibrated Newton constant, Standard Model embedding, or universal Braid-only uniqueness is claimed.

Abstract

We develop Subjectivity-Intersection Braid Quantum Gravity as a candidate theory in which a public spacetime carrier is generated by intersections of nonidentical, standpoint-bearing Braid processes rather than assumed as an ontologically prior container. Subjectivity here denotes a located mode of access, transport, or presentation, not consciousness. The construction begins with one pointed quantum-algebraic Braid source and maintains an explicit separation between finite source theorem, continuum limit, physical interpretation, and empirical claim.

The fixed source produces a permutation-covariant rank-four event moment

$$Q_4 = \frac{12}{25}P_0 + \frac{52}{25}P_1$$

and a source-null causal reflection $J_{\text{ray}} = -P_0 + P_1$. Their product $K_{\text{evt}} = J_{\text{ray}}Q_4$ has exact signature $(3+, 1-)$. The same marked event structure selects the full matter corner, ten independent response directions, fixed-model matter routing, and the relative normalization 3/5 without manual insertion. A refinement clock fixes the Euclidean event-semigroup rate $r = 2\pi/3$ and selects the Lorentzian infrared principal tensor by source continuation rather than a Wick-sign declaration.

The post-v0.7 advance is a controlled passage from this carrier to a continuous tensor and then to sourced dynamics. Eight typed source sectors carry exact rational $GL(4)$ transport and finite nonmetricity. Braid-derived Gibbs tangents extend them to a smooth event cylinder; the metric, its tangent, and nonmetricity glue over all 24 source charts and transform covariantly under passive local $GL(4)$. Event-path Fisher/Bogoliubov–Kubo–Mori horizontality uniquely lifts metric change to finite positive Markov rates. The lift is flat on all 45 metric two-planes and integrates locally to an exponential leaf. A source-product typing condition then removes the remaining solder ambiguity. Finally, the co-moving causal reflection

$$J(Q, \tau) = I - 2 \frac{(Q\tau)\tau^\top}{\tau^\top Q \tau}, \quad K(Q, \tau) = Q - 2 \frac{(Q\tau)(Q\tau)^\top}{\tau^\top Q \tau}$$

restores all ten Lorentz-metric variations without introducing an independent clock coupling. In the declared long-wavelength, local, second-order, formally self-adjoint, conservative class, the finite Braid action variation converges to Hilbert variation, yields an off-shell Noether identity and on-shell Ward conservation, and combines with the source-native first-order gravity sector to give

$$G_{\mu\nu}(g) = \frac{3}{5}T_{\mu\nu}(g, \lambda).$$

Its flat quadratic limit is the massless Fierz–Pauli gauge class with two physical helicities in four dimensions.

The theorem is scoped. The finite-to-continuum residual is proved only as $O(h)$, although every odd midpoint Bregman term cancels exactly. The unmodified full finite Umegaki/Jeffreys action fails the tested one-link reflection-positivity condition. A source-native GKSL evolution admits an exact right-tail Stinespring realization and an exact conserved total temporal charge, but the three tested spatial defects lie outside the environment-only charge range. Thus full finite Lorentzian backreaction and four-component discrete total Ward balance remain open. The results refute the universal necessity thesis of *objectivist primacy* by construction; they do not refute mind-independent reality, establish phenomenal consciousness, or show that nature realizes the model.

Keywords: Subjectivity-Intersection Braid Quantum Gravity; quantum gravity candidate; pointed braid; generated objectivity; objectivist primacy; Lorentz signature; emergent spacetime; Einstein equation; metric-affine gravity; Ward identity.

日本語要旨

本稿は、主観交差Braidを根底とする量子重力候補のフル版v0.8を提示する。ここで「主観」は意識ではなく、アクセス・輸送・表示の位置づけられた様態を意味する。異なる立場を消去せず、その交差から共有可能なLorentz時空carrierが生成される、という存在論的順序を有限source模型として実装する。

v0.7までに、Braid event moment Q_4 、source因果反転 J_{ray} 、Lorentz carrier $K_{\text{evt}} = J_{\text{ray}} Q_4$ 、固定模型内のMMR2解除、十個の物質応答、および手動設定でない相対係数3/5が得られていた。v0.8では、このcarrierを作用、連続テンソル、stress/Ward、backreactionへ接続した。refinement clockはrate $2\pi/3$ を固定し、source continuationはIR Lorentz principal tensorを選ぶ。八つのsource sector上の有限 $GL(4)$ 輸送とnonmetricityは、Gibbs変形、event-cylinder chain rule、全24 chartのgluing、local $GL(4)$ 自然性を経て連続tensor fieldとなる。BKM水平条件は計量変化に対応する有限Markov rateを一意にし、その接続は45/45の二平面でflat、有限指数leafへ積分可能である。

さらにsource-product型がmetric solderを一意化し、co-moving grading $J(Q, \tau)$ が固定gradingで欠けた時間空間混合三方向を回復する。これにより、宣言した長波長・局所・二階・形式的自己共役・保存的クラス内で、有限Braid作用の変分からHilbert stressとon-shell Ward保存則が導かれ、同じmetricを変分するsource-native重力作用との結合から

$$G_{\mu\nu}(g) = \frac{3}{5} T_{\mu\nu}(g, \lambda)$$

が得られる。flat二次極限は四次元massless spin-2のFierz--Pauli gauge classであり、物理helicityは二つである。

ただし完成量子重力ではない。有限から連続への残差は現状 $O(h)$ 、full finite Umegaki/Jeffreys作用は標準one-link reflection positivityを満たさない。source-native GKSL発展はright-tail環境によるStinespring表示とtotal $G1$ 保存を持つが、三つのspatial defectはenvironment-only charge rangeの外にある。したがってfull finite Lorentzian backreactionと四成分discrete Wardは未完である。客観的Lorentz構造を原始的に置くことがあらゆる構成で必要だという「客観主義的優先性」は構成的に反証されるが、自然界での実現、意識、経験的重力を証明したとは主張しない。

1 What changed from v0.7

Version 0.7 stopped at a source-derived Lorentz carrier and treated the Source-Dynamical Parent Completion (SDPC) as one remaining compound bridge. Version 0.8 does not silently assume that bridge. It decomposes

it into action selection, tensor continuation, active transport, solder uniqueness, Ward conservation, and backreaction gates, and records both the successful derivations and the exact reversals that occurred under red-team audit.

Question	v0.8 result	Remaining boundary
Operational four-direction carrier	Retained exactly from the source cylinder and four null links	No universal four-dimensional uniqueness theorem
Matter realization and 3/5	Fixed-model MMR2 closed; normalized source-cup ratio derived	Arbitrary instruments and SI-unit calibration remain open
Lorentz action selection	Source reflection semigroup and refinement clock select the IR principal tensor	Full finite interacting Lorentzian functional remains open
Continuous tensor field	S4 gluing and passive local $GL(4)$ naturality exact	Empirical identification with natural spacetime remains open
Physical stress and Ward law	Derived in the declared local conservative class	Higher derivative, nonlocal, and full finite dissipative sectors remain open
Einstein and spin-2 limit	Noncircular scoped derivation with two helicities	UV completion and finite quantum backreaction remain open
Finite open-system balance	Exact total temporal charge with Braid right-tail environment	Three spatial environment-only balances obey a range NO-GO

Table 1: The v0.8 dependency split. “Derived” always means derived inside the declared fixed source model and stated continuum class, not established as a universal law of nature.

The strongest defensible claim is consequently upgraded but also narrowed with greater precision. SDPC is no longer carried as an undifferentiated assumption in the declared long-wavelength class: its IR grading, active transport, physical solder, and conservative Ward content have explicit source derivations. The unrestricted SDPC claim is not declared solved, because the full finite Lorentzian functional and the dissipative four-current balance remain incomplete.

2 Subjectivity intersection and the target of the refutation

2.1 Subjectivity is not consciousness

A *subjectivity* in this work is a typed and located mode of access, transport, or presentation whose distinction from other such modes is constitutive of the construction. It may be represented by a source prefix, a reference system, or an oriented transport context. No braid, photon, clock, algebra, or source sector is thereby attributed sensation, awareness, intention, or phenomenal experience. Any bridge from this formal notion to consciousness would require a separate axiom and separate evidence.

Let A and B denote nonidentical source standpoints. Their intersection does not mean a third external observer that views both from nowhere. It is the compositional event in which their difference is retained while presentation redundancy is removed. The public structure $C(A, B)$ is then an output of the intersection rather than an independently completed container.

Definition 2.1 (Generated objectivity). A carrier $C(A, B)$ exhibits generated objectivity when it is invariant under allowed internal presentation changes, covariant under ambient re-description, symmetric under exchange of the two typed standpoint roles when the source law requires it, altered by the deletion of either role, and regenerated by one source law through the declared refinement stages without stage-dependent fitting.

2.2 The exact objectivist claim being denied

Definition 2.2 (Objectivist Primacy Thesis, OP). Every mathematically admissible construction of a stable, shareable, transformation-respecting physical carrier must take that carrier, or an equivalent perspective-neutral carrier, as primitive. Distinct standpoints may describe it, but cannot generate it.

OP is a universal necessity thesis. It is stronger than scientific reproducibility, realism about measured regularities, covariance, or mind-independent existence. A single valid generated-objectivity model refutes OP as stated. It does not prove that the model is realized in nature. The present paper therefore keeps two conclusions separate:

1. a mathematical counterexample to the necessity of objectivist primacy; and
2. a physical quantum-gravity hypothesis whose empirical truth remains open.

The proposal is adjacent to, but not identical with, relational quantum mechanics, quantum-reference-frame programmes, QBism, perspectival realism, and structural realism [11, 5, 14, 3, 10, 8, 15]. Its distinct technical claim is generative: a shared tensor carrier is reconstructed from maintained source provenance and intersection rather than presupposed as a perspective-neutral starting object.

3 Pointed-Braid source and operational event cylinder

3.1 Quantum-algebraic source

At refinement stage n , take the matrix algebra

$$\mathcal{A}_n = M_{5n}(\mathbb{C}), \quad j_n(X) = X \otimes I_5. \quad (1)$$

The source contains distinguished states, marked local algebras, and unitary Braid transports satisfying the typed Yang–Baxter and composition laws. This is quantum in the algebraic sense: states, complex noncommutative observables, tensor composition, and unitary transport are present. The present paper does not derive complex amplitudes, the Born rule, or Hilbert-space composition from an unstructured topological braid group.

The source data are therefore neither “bare braid topology” [1] nor a continuum spacetime. They are a finite quantum-algebraic pointed-Braid source developed in the preceding Subjectivity-Intersection Mathematics programme [16, 17]. The scientific question is how much geometry and dynamics this source forces without target fitting.

3.2 Operational event identification

The fixed source packet contains a marked event effect P and a spectral event algebra. The actual stored event transition R_{evt} acts on 25 local atoms, is involutive, and satisfies the declared Yang–Baxter event relation. The same marked generator determines four future-null affine links. This identification retires an independent formal correspondence principle CGR inside the fixed model: the event carrier is read from the source law itself rather than attached as a separate map.

This is a scoped statement. It does not say that every operational event in nature must be represented by this P , nor does it select a natural signed sector. It says that the previously separate formal event correspondence is redundant in the declared source construction.

3.3 Four dimensions without a primitive completed spacetime

The operational cylinder has four source directions because four independently certified null links span the event carrier. Let L be the matrix with these links as columns. The exact certificate gives

$$\det L = \frac{81}{16} \neq 0. \quad (2)$$

Thus the event carrier is four-dimensional in the fixed model. This is stronger than postulating an arbitrary ambient $\mathbb{R}^4$ and then placing Braid data inside it: the four operational directions are generated and are nondegenerate. It is weaker than a theorem that four dimensions are inevitable for all admissible Braid sources. The latter is not claimed.

4 Matter routing and the non-manual factor 3/5

4.1 The full event-conditioned corner

The marked event defines the physical corner candidate

$$\mathcal{A}_\cap = PAP, \quad (3)$$

with Hamiltonian $H_\cap = PG_1P$ and ten compressed source operators

$$F_A^\cap = PF_A P, \quad A = 1, \dots, 10. \quad (4)$$

The corresponding Gibbs family is faithful on the corner, and its Bogoliubov–Kubo–Mori Hessian has rank ten. Hence the ten declared matter-response directions are locally nondegenerate.

The source atoms generate five partition retractions. Requiring exact preservation of the already fixed eleven-operator packet

$$\mathcal{S}_\cap = \{PG_1P, PF_1P, \dots, PF_{10}P\} \quad (5)$$

selects only the full partition. Every nontrivial dephasing changes $H_\cap$ before any coefficient can be retuned. This closes MMR2 within the declared fixed event-conditioned model and complete five-partition class. It does not prove uniqueness among all completely positive instruments or all possible subalgebras.

4.2 Where 3/5 comes from

The normalized tail trace sends the corner unit to the outer event effect P . Its source-cup composition yields the relative multiplier

$$\kappa_B = \frac{3}{5}. \quad (6)$$

No line of the Lorentz-carrier or backreaction calculation sets or retunes this value. It is inherited from the same marked source routing that selects the event-conditioned matter packet. In particular, the appearance of the same coefficient in the Einstein equation is not counted as proof of that equation: source-derived normalization, tensor typing, conservation, and common metric variation are proved as separate gates.

5 Braid-derived local-label dynamics

5.1 Actual event transition versus identity and flip

For each adjacent pair of source directions, define the displacement under the actual Braid event transition. Its exact pair moment is

$$M_{\text{pair}} = \begin{pmatrix} 28 & -20 \\ -20 & 28 \end{pmatrix}, \quad (7)$$

with eigenvalues 8 and 48 and rank two. The identity comparator has rank zero and the ordinary flip comparator rank one. Thus the rank-two coupling is specific to the actual non-flip event map within this comparator class.

Average the adjacent-pair moment over the 24 source charts. The result is

$$Q_4 = \frac{104I_4 - 201\mathbf{1}^\top}{50} = \frac{12}{25}P_0 + \frac{52}{25}P_1, \quad (8)$$

where

$$P_0 = \frac{1}{4}\mathbf{1}\mathbf{1}^\top, \quad P_1 = I_4 - P_0. \quad (9)$$

Q_4 has diagonal entries $42/25$, off-diagonal entries $-2/5$, and eigenvalues $12/25$ on the uniform line and $52/25$ on its three-dimensional complement. It is positive and full rank. No distinguished physical $1 + 3$ ordering or external axis weights enter Eq. (8).

5.2 Relative entropy and the BKM–Dirichlet form

The source response is the derivative of a log-partition function. Within the response-exact Bregman class this selects the log- Z Bregman divergence, equivalently the Umegaki relative entropy for the Gibbs family [13, 2]. On actual neighbor event edges, the quadratic term is

$$\delta^2 S_{\text{edge}} = \frac{1}{2} \int Q_4^{ab} \langle \partial_a \lambda, \partial_b \lambda \rangle_{\text{BKM}} d\mu. \quad (10)$$

The base moment has rank four and the internal BKM metric rank four in the tested local-label sector, so the gradient quadratic form has rank sixteen. This supplies source-independent local-label dynamics on the fixed cylinder. It is a positive Dirichlet form, not yet a Lorentzian action.

6 Null tetrad and source-derived Lorentz carrier

6.1 The source-null metric

The four null links have a source-fixed Gram matrix G_{null} with zero diagonal and off-diagonal entries $-12/25$. Since L is invertible, the ambient metric reconstructed from the links is unique:

$$g = L^{-\top} G_{\text{null}} L^{-1} = \text{diag}\left(-\frac{16}{25}, \frac{16}{75}, \frac{16}{75}, \frac{16}{75}\right). \quad (11)$$

It has signature $(3+, 1-)$. The induced map on symmetric tensors, $\text{Sym}^2(L)$, has rank ten. Thus the four-direction carrier is algebraically capable of supporting all ten symmetric rank-two components.

This statement alone does not solve the physical stress problem. Full rank is an existence fact about an algebraic map; it does not prove that the actual matter operator packet transforms through that map or that the resulting response is conserved.

6.2 Why a coframe alone cannot create the Lorentz sign

The positive event moment remains positive under the null coframe:

$$LQ_4L^\top = \text{diag}\left(\frac{27}{25}, \frac{117}{25}, \frac{117}{25}, \frac{117}{25}\right). \quad (12)$$

By Sylvester’s law of inertia, no real invertible congruence changes signature $(4+, 0-)$ to $(3+, 1-)$. Therefore a claim that “the null tetrad automatically Lorentzianizes the positive edge action” is false. This no-go forces the causal sign to have a source origin other than coordinate change.

6.3 Causal grading

The source causal analysis distinguishes the uniform sum of the four future-null rays as the timelike direction. The corresponding reflection is

$$J_{\text{ray}} = I_4 - \frac{1}{2}\mathbf{1}\mathbf{1}^\top = -P_0 + P_1. \quad (13)$$

It has eigenvalue -1 on the uniform line and $+1$ on the standard three-dimensional complement. Since Q_4 is S_4 -equivariant, J_{ray} and Q_4 commute. Hence

$$K_{\text{evt}} = J_{\text{ray}}Q_4 = -\frac{12}{25}P_0 + \frac{52}{25}P_1 \quad (14)$$

is symmetric with exact signature $(3+, 1-)$. In components its diagonal is $36/25$ and every off-diagonal entry is $-16/25$.

Theorem 6.1 (Source-derived Lorentz event carrier). *In the declared fixed source model, the actual Braid event moment Q_4 and the independently source-selected causal grading J_{ray} determine the Lorentz-signature carrier $K_{\text{evt}} = J_{\text{ray}}Q_4$. The construction uses no Wick rotation, manual sign assignment, or fitted continuous coefficient.*

6.4 Natural normalization to the null Gram tensor

Because Eq. (14) is contravariant in the kinetic form, its physical comparison target is G_{null}^{-1} , not G_{null} . Restrict a normalizer to the positive, self-adjoint, S_4 -equivariant class

$$A_{\text{kin}} = a_0P_0 + a_1P_1. \quad (15)$$

Then

$$A_{\text{kin}}^\top K_{\text{evt}} A_{\text{kin}} = G_{\text{null}}^{-1} \quad (16)$$

fixes

$$a_0 = \frac{25\sqrt{3}}{36}, \quad a_1 = \frac{25\sqrt{39}}{156} \quad (17)$$

uniquely within that class. All 24 chart permutations commute with the projector decomposition. In the character frame one recovers

$$g^{-1} = \text{diag}\left(-\frac{25}{16}, \frac{75}{16}, \frac{75}{16}, \frac{75}{16}\right). \quad (18)$$

The uniqueness claim is deliberately local: it is not uniqueness under arbitrary Lorentz isometries or among all nonlinear field redefinitions.

7 From a Lorentz carrier to a Lorentz action

7.1 Why the action question is separate

The positive Umegaki edge functional underlying Eq. (10) has Hessian $Q_4 \otimes G_{\text{BKM}}$. The causal carrier has principal tensor $K_{\text{evt}} \otimes G_{\text{BKM}}$. Since the latter contains one negative mode, equality of the two cannot be obtained by a real invertible coframe congruence. A carrier identity is therefore not yet an action theorem.

Three tempting shortcuts fail exactly. First, the canonical Hamiltonian-orbit action generated by G_1 is first order in velocity and has zero velocity Hessian. Its existing metric response has rank three, not four. Second, forward and reverse event displacements have the same second moment; signed orientation averages vanish and scalar combinations remain scalar multiples of Q_4 . Third, the doubled closed-time-path form

$$\eta_{\text{br}} \otimes Q_4, \quad \eta_{\text{br}} = \text{diag}(1, -1), \quad (19)$$

has signature $(4+, 4-)$ on branch $\otimes$ event space, while any fixed branch contraction is again only a scalar multiple of Q_4 . None selects the one-dimensional clock line inside the four-dimensional event carrier.

A graded trace does perform the algebraic conversion,

$$\text{Str}_J(X) = \text{Tr}(J_{\text{ray}}X), \quad \text{Str}_J(Q_4) = J_{\text{ray}}Q_4, \quad (20)$$

and is unique up to scale among graded-cyclic traces fixed by this $\mathbb{Z}_2$ grading. It is not positive because $\text{Str}_J(P_0) = -1$. Declaring a change from ordinary positive operator algebra to a graded category would therefore make a conditional construction, not a Braid-only derivation.

7.2 Positive causal sectors and source continuation

The same Lorentz Hessian can be written as a difference of ordinary positive sector Hessians,

$$H_0 = \frac{12}{25}P_0 \otimes G_{\text{BKM}}, \quad H_1 = \frac{52}{25}P_1 \otimes G_{\text{BKM}}, \quad H_1 - H_0 = K_{\text{evt}} \otimes G_{\text{BKM}}. \quad (21)$$

Its signature is $(12+, 4-)$ when the internal BKM tangent has rank four. This removes the need for a nonpositive trace but still leaves a continuation rule between the positive source functional and the causal difference.

The source causal reflection satisfies

$$J_{\text{ray}}^T Q_4 J_{\text{ray}} = Q_4, \quad J_{\text{ray}}^T Q_4^{-1} J_{\text{ray}} = Q_4^{-1}. \quad (22)$$

The raw 25-event involution P consists of nine fixed points and eight transpositions. Hence P has eight eigenvalues -1 and cannot be represented as e^{-aH} with $H \geq 0$. The positive generator is instead the graph Laplacian

$$L_{\text{evt}} = I - P, \quad K_t = e^{-rtL_{\text{evt}}}. \quad (23)$$

The marked source refinement interval $\Delta\tau = 3/4$ and the minimal Stone endpoint π fix

$$r = \frac{2\pi}{3}. \quad (24)$$

There is therefore no fitted event rate. Reflection continuation of the source clock line changes the uniform sector sign and leaves the three-dimensional complement positive.

Theorem 7.1 (Scoped source selection of the infrared Lorentz action). *Within the declared real Stone/affine completion of the fixed marked source, the event semigroup with rate (24), the source reflection, and the response-exact BKM edge functional select the infrared quadratic tensor*

$$K_{\text{evt}} \otimes G_{\text{BKM}}. \quad (25)$$

No manual Wick sign, external rate, or continuous coefficient fit is required.

The theorem is an infrared principal-symbol theorem. It does not make the unmodified full finite Umegaki functional Lorentzian, a distinction that becomes decisive in Section 13.

8 Quantum channel and finite metric-affine bundle

8.1 A source-native quantum channel on the same carrier

For each of the eight source masks, let R_m denote the normalized ordered-edge Gram matrix. Each R_m is exactly positive definite with unit diagonal. The Schur map

$$\Phi_m(X) = R_m \circ X \quad (26)$$

is consequently completely positive, trace preserving, and unital. It is nontrivial and admits a Stinespring representation. The source-null tetrad normalized by its source norm gives

$$U = \frac{2}{3}L = \frac{1}{2}H_4, \quad (27)$$

an orthogonal Hadamard transformation. It sends the character clock projector to the ray clock projector and intertwines the source channel with

$$P_0, \quad J_{\text{ray}}, \quad Q_4, \quad K_{\text{evt}}. \quad (28)$$

Thus a standard physical quantum operation and the Lorentz causal carrier live on one typed four-carrier rather than on separately postulated state spaces.

This result does not identify a derivative. The eight endpoint Grams are distinct while the anchor K_{evt} is common. Linear and quadratic positive paths joining the same endpoints have different initial derivatives. Endpoints alone cannot select a tangent, Hilbert stress, or Ward law.

8.2 Direct rational carrier transport

Let C_m be the unique four-column carrier matrix in source sector m . The source-selected least-squares transport is

$$T_{m \leftarrow n} = (C_m^\top C_m)^{-1} C_m^\top C_n. \quad (29)$$

Every one of the 48 oriented source-cube packets is rank four and therefore invertible. The ordered-edge Gram is not preserved:

$$N_{mn} = T_{m \leftarrow n}^\top G_m T_{m \leftarrow n} - G_n. \quad (30)$$

All 48 defects are symmetric and rank four at both fixed good primes. This is a finite metric-affine structure: connection and nonmetricity are outputs of the same source.

Equation (30) must not be divided directly by the spacetime mesh. Its endpoints are internal source sectors, not neighboring spacetime cells. The type error was explicitly tested and rejected. Continuous differentiation instead comes from the already existing Gibbs source coordinates.

8.3 Characteristic-zero tangent theorem

Let θ_i , $i = 1, 2, 3$, denote the source-local Gibbs labels. Differentiating the carrier module and the ordered-edge Gram along these labels yields

$$\dot{C}_{m,i} = 0, \quad A_{m,i} = 0, \quad N_{m,i} = \dot{G}_{m,i}. \quad (31)$$

The equalities for the carrier and its projection connection were checked as exact rational identities: 288 of 288 carrier commutators vanish across eight typed-gauge-inequivalent sectors and three label directions. In contrast, every one of the 24 Gram tangents is nonzero, symmetric, and rank two. Hence

$$N_{m,i} = \dot{G}_{m,i} \neq 0, \quad \text{rank } N_{m,i} = 2. \quad (32)$$

The metric changes while the preferred source gauge keeps the carrier stationary to first order. In a nonconstant frame the connection is, of course, not zero; the inhomogeneous frame term restores the usual tensor law.

9 Continuous event-cylinder tensor field

9.1 Chain rule without an external ambient manifold

The smooth event cylinder was defined before the metric-continuation calculation. Its local label fields are not added for this theorem. Across all eight sectors the three metric responses have rank three. Combining them with the four-dimensional event base gives the rank-twelve chain-rule symbol

$$\partial_a G_m(x) = \sum_{i=1}^3 \frac{\partial G_m}{\partial \theta_i} \partial_a \theta_i(x), \quad \text{rank } \sigma(\partial G) = 4 \times 3 = 12. \quad (33)$$

Equation (33) is the first-order passage from finite source labels to a continuous local field. It does not identify the cylinder with measured natural spacetime; it establishes a source-derived operational base on which differentiation is defined.

9.2 Chart gluing and local $GL(4)$ naturality

The 24 source charts act by the four-ray permutation representation. Metric, metric tangent, and nonmetricity obey the same congruence law in every chart. Exact checks include 192 metric packets, 576 tangent/nonmetricity packets, and 13,824 double-overlap compositions. Direct $A \rightarrow C$ transport equals $A \rightarrow B \rightarrow C$ transport.

The finite result extends to arbitrary passive local frame changes. With the source-fixed solder U , a frame Jacobian $J(x)$ acts on the carrier by

$$C_J(x) = UJ(x)U^\top. \quad (34)$$

Writing A for the source-gauge connection, the transformed objects are

$$G' = C_J^\top G C_J, \quad (35)$$

$$A' = C_J^{-1} A C_J + C_J^{-1} dC_J, \quad (36)$$

$$N' = dG' - A'^\top G' - G' A' = C_J^\top N C_J. \quad (37)$$

The representation product, its S_4 restriction, the actual metric, the variable-frame connection, and nonmetricity all pass exact composition tests. Thus the result is not merely 24 coincident matrices: it is a local tensor field on the declared event domain.

Theorem 9.1 (Source-event tensor gluing). *The Gibbs-derived ordered-edge metric and its nonmetricity define an S_4 -glued, passively local- $GL(4)$ -natural first-order tensor field on the smooth source event cylinder. No external $\mathbb{R}^4$ coordinate field or fitted gluing coefficient is required.*

Passive covariance alone does not select a matter action. Many curvature-improved actions are covariant and conserved while differing in their Hilbert stress. The next step therefore asks whether the source Markov law fixes active metric dependence.

10 Active Braid transport and the ten metric directions

10.1 Minimal conservative operator

The source cell measure multiplied by the metric volume density transforms naturally under local $GL(4)$. For a local second-order scalar operator with the derived principal symbol and density, formal self-adjointness fixes the four drift coefficients. Markov conservativity fixes the zero-order potential to zero. In particular, an

independently added αR improvement is excluded within this declared class because it violates preservation of the constant function.

This is a uniqueness statement within a class, not a theorem that all physical actions must be second order or Markovian. The critical remaining issue is whether a change in the derived metric uniquely changes the actual Braid transition rates.

10.2 Horizontal conductance lift

There are eight nontrivial undirected Braid-event edges and twelve source-typed conductance modes. Their moment map to symmetric metric components has rank ten and a two-dimensional kernel. Every one of the 24 actual metric tangents lies in its image, but the metric alone cannot see the two residual edge coordinates.

The event-path probability law supplies the missing criterion. Equip rate variations with the Fisher/BKM pairing at the source rate $2\pi/3$. Among all rate variations producing a given metric variation, select the vector orthogonal to the kernel. The raw 48-variable kernel has dimension 38 and full BKM rank 38; the residual two-dimensional typed kernel also has BKM rank two. The horizontal lift is therefore unique. It reproduces 24 of 24 source tangents, is S_4 -covariant, and fixes the common-to-relative quotient weight ratio to 18 : 1 without fitting.

Let M denote the rank-ten moment matrix and r_0 the source rate vector. The horizontal leaf has the finite form

$$r(\xi) = r_0 \circ \exp(M^T \xi), \quad (38)$$

where the exponential and product are componentwise. Its Jacobian is

$$DM(\xi) = M \operatorname{diag}(r(\xi)) M^T > 0. \quad (39)$$

Hence the metric moment is locally invertible, rates remain positive, and row-sum zero, detailed balance, and refinement compatibility are preserved. The horizontal connection has exact zero curvature on all 45 independent metric two-planes. Equation (38) is consequently a local finite integration, not merely a first-order fit.

10.3 The causal 7 + 3 split

The positive X21R1 Gram is not itself the physical Lorentz metric: neither it nor its inverse matches the positive event moment in any of the eight sectors. The event sector at fixed causal reflection supplies seven J -even metric directions. The three Gibbs-label directions supply precisely the missing J -odd time-space directions. Their combined response has rank ten in all sectors and glues as a tensor.

At this point a subtle variational obstruction appears. If the odd metric variables are identified with the matter labels θ_i , holding matter fixed during a Hilbert variation locks those three directions and leaves rank seven. The remedy cannot be to vary matter while claiming a fixed-matter stress. The source already contains ten matter-independent source variables a_A , but covariance alone is insufficient to solder them: the S_4 commutant on Sym^2 has dimension nine, and clock, grading, and event spectrum still leave a five-dimensional family.

The actual source typing removes this ambiguity. The ten variables are not an arbitrary ten-dimensional representation; they are generated as symmetric products of four carrier variables. Requiring preservation of that product generation and of the already derived four-dimensional map U forces

$$A \longmapsto U^T A U \quad (40)$$

with no residual continuous freedom. This is the physical solder used below.

Proposition 10.1 (Typed solder uniqueness). *Within the existing source-product class, Eq. (40) is the unique rank-ten solder compatible with the four-carrier map, source typing, and chart action. Covariance without the product-typing condition is not sufficient.*

11 Co-moving causal grading, Hilbert stress, and Ward identity

11.1 The fixed-grading obstruction

A fixed J_{ray} transports only seven of the ten positive-form variations into symmetric Lorentz-metric variations. This caused an explicit reversal of an earlier stress/Ward completion claim. The missing three directions are the time–space mixtures. They are recovered only if the clock projector moves with the positive event form.

Let τ be the source clock ray, normalized only projectively. Define

$$J(Q, \tau) = I - 2 \frac{(Q\tau)\tau^\top}{\tau^\top Q \tau}, \quad (41)$$

$$K(Q, \tau) = J(Q, \tau)Q = Q - 2 \frac{(Q\tau)(Q\tau)^\top}{\tau^\top Q \tau}. \quad (42)$$

At the source anchor these recover J_{ray} and K_{evt} exactly. The off-shell Jacobian $D_Q K$ has rank ten. The clock derivative has rank three but is not an independent coupling:

$$D_\tau K = (D_Q K) C, \quad (43)$$

for an exact source-derived coefficient map C , while the clock-scale direction lies in the kernel. Moreover, the clock reflection is self-inverse on the tested positive-form domain,

$$K = \mathcal{R}_\tau(Q), \quad Q = \mathcal{R}_\tau(K). \quad (44)$$

Thus the clock is a redundant local parametrization of the full-rank Lorentz tensor, not a new continuum field inserted to restore rank.

11.2 Continuum matter action and stress

Let λ^A denote the matter coordinates and $\mathbf{G}_{AB}(\lambda)$ the BKM metric inherited from the full source corner. The source-selected long-wavelength action in the declared class is

$$S_m[g, \lambda] = \frac{1}{2} \int_{\mathcal{M}} d^4x \sqrt{-g} g^{\mu\nu} \mathbf{G}_{AB}(\lambda) \partial_\mu \lambda^A \partial_\nu \lambda^B. \quad (45)$$

This is not an independently guessed nonlinear sigma model. Its local density, principal tensor, internal BKM pairing, edge coupling, active rate dependence, and metric solder are the continuum images of the finite source construction described above.

Variation through the independent source variables a_A and the unique solder gives the Hilbert tensor

$$T_{\mu\nu} = \mathbf{G}_{AB} \left(\partial_\mu \lambda^A \partial_\nu \lambda^B - \frac{1}{2} g_{\mu\nu} g^{\rho\sigma} \partial_\rho \lambda^A \partial_\sigma \lambda^B \right). \quad (46)$$

Diagonal diffeomorphism covariance yields the off-shell identity

$$\nabla^\mu T_{\mu\nu} = -\mathcal{E}_A(\lambda) \partial_\nu \lambda^A, \quad (47)$$

where $\mathcal{E}_A = 0$ are the matter Euler–Lagrange equations. Therefore

$$\mathcal{E}_A = 0 \implies \nabla^\mu T_{\mu\nu} = 0. \quad (48)$$

Theorem 11.1 (Scoped Braid-source stress/Ward theorem). *For the fixed source, and within the long-wavelength, local, second-order, formally self-adjoint, Markov-conservative class, the finite response-exact Braid action, horizontal active transport, source-product solder, and co-moving causal reflection determine Eq. (45). Its fixed-matter variation gives Eq. (46), and its source symmetry gives the off-shell and on-shell Ward laws (47).*

The restrictions in Theorem 11.1 are noncompensating. The theorem does not cover arbitrary curvature improvements, higher derivatives, nonlocal actions, or the full finite dissipative quantum evolution. Within its declared class, however, no separate MMR, stress solder, Ward postulate, causal-sign insertion, or coefficient fit remains.

12 Noncircular backreaction and the Einstein limit

12.1 Common metric variation

The gravity and matter sectors are varied with respect to the same source-derived metric. In the long-wavelength, low-curvature, vanishing-hypermomentum sector, the source-native first-order gravity functional takes the metric-affine form

$$S_{\text{IR}}[g, \Gamma, \lambda] = \frac{1}{2\kappa_B} \int d^4x \sqrt{-g} g^{\mu\nu} R_{(\mu\nu)}(\Gamma) + S_m[g, \lambda], \quad \kappa_B = \frac{3}{5}. \quad (49)$$

The value $3/5$ is the normalized outer-cup routing ratio derived before the metric variation. It is not selected by matching an Einstein target. The compatible connection equation removes the projective mode and gives $\Gamma = \Gamma_{\text{LC}}(g)$ in this sector. Metric variation therefore gives

$$\boxed{G_{\mu\nu}(g) = \frac{3}{5} T_{\mu\nu}(g, \lambda)}. \quad (50)$$

The Bianchi identity is consistent with Theorem 11.1; neither side is assigned a conservation law independently to force agreement.

Theorem 12.1 (Scoped noncircular Einstein backreaction). *In the fixed pointed-Braid model and the continuum class of Theorem 11.1, the source-native first-order gravity sector and full-corner matter sector share the same metric variation. Their Euler–Lagrange equation is Eq. (50). No new Einstein–Hilbert term, mixed fitting term, manual $3/5$, or independent stress/Ward axiom is introduced at this stage.*

This is the precise sense in which the previous conditional SDPC bridge is removed in v0.8. It is not a theorem over every imaginable Braid model or every continuum action. It is an internally derived result in one fixed source model and one explicitly delimited universality class.

12.2 Low-energy spin two

Linearize $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ about the flat source anchor. The quadratic gravity action belongs to the massless Fierz–Pauli gauge class [4],

$$\mathcal{L}_{\text{FP}} = -\frac{1}{2} \partial_\lambda h_{\mu\nu} \partial^\lambda h^{\mu\nu} + \partial_\mu h^{\mu\nu} \partial^\lambda h_{\lambda\nu} - \partial_\mu h^{\mu\nu} \partial_\nu h + \frac{1}{2} \partial_\lambda h \partial^\lambda h, \quad (51)$$

with gauge symmetry $\delta h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Coupling to the conserved tensor (46) preserves the linear gauge constraint. In four dimensions the physical massless sector has two helicities.

The claim is a low-energy sector statement. It does not quantize the interacting continuum metric, establish a graviton Fock space on arbitrary backgrounds, or prove perturbative or nonperturbative ultraviolet completion.

12.3 Finite-to-continuum residual

Let $h = 5^{-n}$ be the declared refinement scale. The first variation of the finite action converges weakly to Theorem 12.1 with the proved bound

$$G_{\mu\nu} - \frac{3}{5}T_{\mu\nu} = O(h). \quad (52)$$

The gravity contribution is bounded by $O(h^2)$, while the current matter regularity proof limits the combined estimate to $O(h)$. Each five-adic refinement therefore decreases the stated upper bound by at least a factor 1/5.

The actual event involution already pairs every directed edge with its reverse. The local matter functional is consequently the Jeffreys pair

$$D(\rho_x \parallel \rho_y) + D(\rho_y \parallel \rho_x). \quad (53)$$

All odd terms in its midpoint Bregman expansion, including the cubic term, cancel exactly. This is not a post hoc symmetrization. It removes the orientation-asymmetry coefficient but does not improve the global bound (52) without stronger inter-cell regularity. An explicit analytic positive-rate counterexample with three log rates (0, 0, 1) shows that internal Markov analyticity alone does not match derivatives across adjacent spacetime cells.

13 Full finite quantum evolution and its present boundary

13.1 Reflection-positivity obstruction

The unmodified full finite Umegaki/Jeffreys functional does not satisfy the tested standard positive one-link transfer condition. Inside the actual 125-dimensional source charge family, the fixed packet $\mu = (-1, 2, 6)$ at source coupling $\alpha = 1/50$ gives a cross-link 3×3 determinant

$$\det K_{\text{link}} = -0.008747721561 \dots < 0. \quad (54)$$

This is a direct reflection-positivity NO-GO for that transfer route, not a disproof of the infrared theorem. A BKM heat kernel exists on the local-label manifold but is not a full spacetime-slice transfer matrix. The involution-doubled block is positive semidefinite, yet has 25 zero modes in dimension 50 and is an idempotent projector rather than finite Hamiltonian evolution.

13.2 Source-native GKSL evolution

The fixed source independently supplies a completely positive trace-preserving semigroup of Gorini–Kossakowski–Sudarshan–Lindblad type [7, 9]. Fixing the event clock rate (24) does not determine the three remaining Hamiltonian controls; distinct generators share the same clock. More importantly, trace preservation is probability conservation, not the gravitational Ward law. An open-system balance must include the environment charge.

Let $\{U_g\}_{g=1}^6$ be the six real Braid actions in the Reynolds channel and let

$$p = \left(\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8} \right). \quad (55)$$

The isometry

$$V\psi = \sum_{g=1}^6 \sqrt{p_g} U_g \psi \otimes |g\rangle_E \quad (56)$$

gives an exact Stinespring dilation at each collision [12]. A finite fixed environment cannot reproduce the entire exponential relaxation autonomously for all time. The native Braid right tail solves the capacity problem: each three-strand block decomposes as

$$125 = 17 \cdot 1 + 28 \cdot 3 + 4 \cdot 6, \quad (57)$$

so it contains four regular six-state registers, and the infinite tail supplies fresh collision ancillas.

The pointed and neutral tail states do not select a regular copy or basepoint. This apparent selector is gauge for the reduced channel. Across eight source sectors and four copies, all marked blocks are related by diagonal-sign environment unitaries that preserve the relevant projector. Changing copy or basepoint changes V only by an environment unitary and leaves the partial-trace channel invariant.

13.3 Exact temporal total balance

The system-only G_1 defect has exact Hilbert–Schmidt norm squared 18 in every sector. It is not a total nonconservation. There exists a six-state environment charge H_E such that

$$V^\dagger(G_1 \otimes I_E + I \otimes H_E)V = G_1 \quad (58)$$

in all eight sectors. The equation is covariant under the environment torsor gauge. For N fresh collisions it telescopes to

$$V_N^\dagger \left(G_1^{(S)} + \sum_{k=1}^N H_{E,k} \right) V_N = G_1^{(S)}|_{t=0}. \quad (59)$$

Thus the dissipative temporal defect is exactly a charge transfer into the Braid tail.

13.4 Spatial range NO-GO

For the three tested spatial source candidates

$$H_i = -i[G_1, P_{\chi_i}], \quad i = 1, 2, 3, \quad (60)$$

the channel defects are nonzero in every sector, with squared norms

$$(24, 14, 14). \quad (61)$$

They are exactly orthogonal to the entire six-state environment-only compression range

$$\text{span}\{U_g^\dagger U_h : g, h = 1, \dots, 6\}. \quad (62)$$

No environment-only charge can cancel them. The temporal balance (58) remains exact, but a four-component discrete Ward identity requires either source-derived system–environment interaction stress or a different Noether momentum. This is the present shortest finite-quantum completion gate.

14 Why subjectivity intersection is operative

The construction is not obtained from braid relations alone. It uses a pointed event effect, nonidentical typed source roles, retained transport provenance, a marked causal clock, and a rule for which intersection output is public. These data do physical work:

1. the event effect selects the operational corner;
2. deletion or collapse of a standpoint changes the source packet;

3. the marked event involution supplies the neighbor law and reverse-edge pairing;
4. the clock line selects the causal reflection and its co-moving extension;
5. the source-product typing makes the ten-component solder unique;
6. the right tail supplies an endogenous environment for reduced quantum evolution.

An unpointed abstract braid group does not provide these roles. An alternative theory could append equivalent marks, states, and provenance; doing so would recreate the operative subjectivity-intersection structure under different terminology.

The philosophical interpretation is not forced uniquely by the algebra. The same structure may be described as marked relational quantum events or provenance-preserving quantum reference data. The SIEL interpretation is that a standpoint is a typed mode of access and that objectivity is what remains stable under lawful intersection and re-description. Nothing in this statement attributes consciousness to a braid, source sector, clock, or quantum channel.

15 Constructive refutation of objectivist primacy

The Objectivist Primacy Thesis **OP** is universal: it says that every stable, shareable, transformation-respecting carrier must be primitive or equivalent to a primitive perspective-neutral carrier. The present construction has the order

$$\begin{aligned}
 &\text{nonidentical pointed source standpoints} \\
 &\quad \longrightarrow \text{marked event, matter corner, and quantum channel} \\
 &\quad \longrightarrow \text{four null directions, causal reflection, and Lorentz carrier} \\
 &\quad \longrightarrow \text{glued tensor, conserved stress, and scoped backreaction.}
 \end{aligned} \tag{63}$$

The public carrier is not present as a completed background at the first step. It is reconstructed, chart-glued, and dynamically coupled while retaining the provenance that generated it. Identity and flip comparators fail essential rank properties, while deletion of the pointing removes the selections used by the construction.

Therefore **OP** is false as a necessity thesis: at least one exact model generates a stable Lorentzian public carrier from nonidentical standpoint-bearing source processes. This conclusion does not deny objective regularity or mind-independent existence. It denies only that ontological priority of a completed objective carrier is universally necessary. The stronger claim that nature realizes this construction remains empirical and is not established here.

16 How complete is the quantum-gravity theory?

The theory now satisfies several requirements that a mere analogy does not:

1. a finite noncommutative quantum source with unitary Braid transport and CPTP channels;
2. a source-derived four-dimensional Lorentz carrier and local tensor field;
3. matter and geometry read from one marked source rather than independently normalized models;
4. a selected long-wavelength action, Hilbert stress, and on-shell Ward identity in a declared class;
5. noncircular Einstein backreaction and massless spin-2 recovery in that class;
6. a finite open-system dilation with exact temporal total-charge transfer.

It is not a completed quantum-gravity theory because the following remain open:

- a full finite interacting Lorentzian parent functional with a positive physical reconstruction;
- a four-component discrete total Ward balance, including interaction stress;
- quantization of the continuum gravitational degrees of freedom beyond the finite source and low-energy quadratic sector;

- an ultraviolet completion or controlled all-scale continuum limit;
- anomaly-free nonlinear constraint and gauge structure beyond the stated class;
- calibrated physical units and Newton constant;
- Standard Model matter content, cosmological/black-hole solutions, and distinct numerical predictions;
- independent replication, peer review, and natural-data validation.

The appropriate description is therefore *a coherent, source-explicit mathematical quantum-gravity candidate with a scoped Braid-derived Einstein limit and an unresolved full finite quantum-gravity completion*. Calling it completed would be premature; calling it only a suggestive analogy would ignore the exact tensor, Ward, backreaction, and balance theorems.

17 Falsifiability and next gates

17.1 Formal falsifiers

The exact package fails under revision-matched replay if any of the following occurs:

- the event involution loses its Yang–Baxter relation, rank-two pair moment, or reverse-edge pairing;
- the event average depends on hidden physical axis weights;
- a nontrivial declared partition preserves the eleven-operator matter packet;
- the source-cup routing fails to yield 3/5 without retuning;
- the null links lose rank four or K_{evt} loses Lorentz signature;
- the refinement clock ceases to fix $2\pi/3$ in the declared completion;
- the rational carrier identities, tensor gluing, horizontal flatness, or typed solder fail exact regeneration;
- the co-moving map loses rank ten or develops an independent clock coupling;
- the finite variation does not converge to the Hilbert variation in the stated class;
- the gravity and matter variations require different metrics or a fitted mixed term;
- the environment gauge changes the reduced channel or invalidates Eq. (58).

17.2 Decisive next theorem

The shortest remaining theoretical gate is not a coefficient search. It is an interaction-stress theorem for the repeated-collision dilation. One must derive, from the same collision isometry, operators Π_i^{int} satisfying a four-component balance of the form

$$V^\dagger \left(P_\alpha^{(S)} \otimes I + I \otimes P_\alpha^{(E)} + \Pi_\alpha^{\text{int}} \right) V = P_\alpha^{(S)}, \quad \alpha = 0, 1, 2, 3, \quad (64)$$

with $\Pi_0^{\text{int}} = 0$ compatible with Eq. (58). The three spatial defects are orthogonal to the environment-only range, so any successful construction must use genuine system–environment cross terms or replace the tested commutator momenta by a source-derived Noether family. A hand-chosen cancelling operator would not count.

The empirical gate remains separate. The theory needs a preregistered discriminator with units and uncertainty against general relativity plus ordinary quantum matter and against lower-commitment relational models. No natural-data confirmation is claimed in v0.8.

18 Claim ledger

Statement	Status	Exact scope
Operational event cylinder and formal CGR retirement	Exact derived	Fixed marked source
Full matter corner and rank-ten response	Exact derived	Fixed event-conditioned corner
MMR2 routing	Closed	Complete five-partition retraction class; not arbitrary instruments
Relative factor $3/5$	Source-derived	Normalized outer-cup routing; no SI calibration
Rank-four event moment Q_4	Exact derived	24-chart source average
Four-null-link Lorentz carrier	Exact derived	Fixed source; no universal dimension theorem
IR causal action tensor	Exact scoped theorem	Real Stone/affine completion and source reflection semigroup
CPTP quantum channel on causal carrier	Exact derived	Eight ordered-edge Gram sectors
Finite $GL(4)$ transport/nonmetricity	Exact derived	48 source-cube packets
Continuous metric/nonmetricity tensor	Exact scoped theorem	Smooth source cylinder; passive local $GL(4)$
Active finite Markov transport	Exact local theorem	BKM-horizontal exponential leaf
Physical metric solder	Unique in class	Source-product-generated Sym^2 typing
Hilbert stress and Ward identity	Final scoped pass	Long-wavelength, local, second-order, self-adjoint, conservative class
Einstein equation $G = (3/5)T$	Final scoped pass	Same class; low curvature and zero hypermomentum
Massless spin-2 with two helicities	Derived IR output	Flat four-dimensional quadratic limit
Finite-to-continuum residual	$O(h)$ bound	Odd midpoint terms cancel; $O(h^2)$ not proved globally
Full finite one-link reflection positivity	NO-GO	Unmodified tested Umegaki/Jeffreys kernel
Source-native GKSL evolution	Exact derived	Reduced finite quantum semigroup
Temporal total G_1 balance	Exact derived	Six-state Braid tail environment; all eight sectors
Spatial environment-only balances	NO-GO	Three tested commutator momenta
Full finite Lorentzian quantum backreaction	Open	Interaction stress and physical reconstruction missing
Completed quantum gravity	Not established	UV, all-scale dynamics, calibration, and validation absent
Constructive refutation of OP	Established in-model	Refutes universal priority, not objective reality
Consciousness or empirical ontology claim	Not made	Subjectivity is not phenomenal consciousness

19 Audit chronology and reversals

The derivation was not monotone. Recording reversals is necessary because a final theorem can otherwise conceal which attractive shortcuts failed.

Gate range	Outcome	Scientific role
BGCE261–265	NO-GO / conditional	Stone velocity Hessian, graded trace, orientation, CTP, and raw transfer do not alone derive the Lorentz action.
BGCE266	PASS	Refinement clock fixes $2\pi/3$ and source continuation selects the IR causal tensor.
BGCE267	NO-GO	Fixed-background principal data do not select a unique improved Hilbert stress.
BGCE268–281	mixed to PASS	Quantum channel/carrier intertwiner, finite connection, characteristic-zero tangent, and rank-two local nonmetricity.
BGCE282–284	PASS / partial	Chain rule, chart gluing, and passive local $GL(4)$ tensor naturality.
BGCE285–289	partial	Minimal conservative operator, conductance lift, horizontal uniqueness, $7 + 3$ completion, and fixed-matter rank-seven obstruction.
BGCE290–291	claimed PASS then reversed	The first stress/Ward claim failed uniqueness and nonlinear-integrability red-team checks.
BGCE292–295	PASS	Flat horizontal connection, exponential leaf, typed solder, and finite-to-Hilbert variation.
BGCE296	reversal	Fixed causal grading loses three Lorentz variations.
BGCE297–299	final scoped PASS	Co-moving grading restores rank ten and removes independent clock coupling; stress/Ward closes in the declared class.
BGCE300R1	full scoped PASS	Noncircular Einstein backreaction and massless spin-2 low-energy sector.
BGCE301R1–303R1	split	$O(h)$ convergence and exact odd cancellation; no global $O(h^2)$ upgrade from Markov analyticity.
BGCE304–305	NO-GO / pivot	Full one-link reflection positivity fails; heat/block routes do not yield finite Hamiltonian evolution.
BGCE306–310	split to PASS	GKSL dilation, native tail, selector gauge removal, and exact temporal total-charge balance.
BGCE311	range NO-GO	Environment-only charges cannot cancel the three tested spatial defects.

20 Conclusion

The v0.8 result is stronger than the v0.7 conditional architecture. The fixed pointed-Braid source now supplies not only a Lorentz carrier, matter routing, and the coefficient $3/5$, but also the infrared causal action selection, a glued continuous metric/nonmetricity tensor, active finite transport, a unique typed physical

solder, Hilbert stress, and on-shell Ward conservation in a precise long-wavelength universality class. The source-native gravity and matter variations then yield $G_{\mu\nu} = (3/5)T_{\mu\nu}$ without an independent SDPC axiom inside that class, and the flat quadratic sector contains the two-helicity massless spin-2 theory.

The result is not promoted beyond its proof. The finite residual is only bounded by $O(h)$. The unmodified full finite action does not possess the tested standard one-link reflection positivity. The source-native GKSL evolution has an endogenous Braid-tail environment and an exact total temporal charge, but the three spatial environment-only balance equations are impossible in the tested range. The next problem is therefore sharply localized: derive interaction stress or the correct source Noether momentum for a four-component discrete Ward identity.

The philosophical conclusion is constructive. Objective Lorentz structure, conserved stress, and a classical gravity limit need not be primitive in every exact model; they can be generated from nonidentical, marked, standpoint-bearing processes and their lawful intersection. Objectivity is retained as stable public invariance. What is rejected is its universal ontological priority. The physical conclusion is correspondingly measured: this is a serious and increasingly complete Braid quantum-gravity candidate, not yet a completed or empirically validated theory.

Data, code, and provenance

The exact certificates and replay scripts are stored in the SIEL Research Agent repository, [SIEL-Research/SIEL-Research-Agent](#). Version 0.8 retains the verified lineage through BGCE260 and adds the revision-matched sequence BGCE261–BGCE311: source continuation and action selection (BGCE261–266), quantum-channel/carrier intertwining and metric-affine transport (BGCE268–281), event-cylinder tensor gluing (BGCE282–284), active Markov transport and solder selection (BGCE285–299), noncircular Einstein/spin-2 recovery and finite residual audit (BGCE300R1–303R1), and finite quantum reconstruction and total-balance tests (BGCE304–311). The source revision immediately before manuscript drafting was commit `d8116552f`. Each central computational claim is linked to a question, source matrix, executable evaluator, raw output or certificate, independent verification record, and Japanese audit report. These are internal exact theoretical and computational certificates, not external replication, peer review, or empirical validation.

Declaration on computational assistance

OpenAI Codex was used under the author’s direction for mathematical ideation, exact-check scripting, audit organization, and drafting. The author retains responsibility for definitions, interpretation, verification, citations, and any decision to release or submit the work.

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